On Monochromatic Orthogonal Adjacencies in the Faces of an $n \times n \times n$ Cube

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Abstract

Consider the six faces of an $n \times n \times n$ cube ($n > 1$), each face divided into an $n \times n$ grid of unit *facelets*, and stickered with exactly six colors, each color used exactly n^2 times. Call two facelets *neighbors* when they are horizontally or vertically adjacent *within the same face* (no diagonal and no across-the-edge adjacencies). We ask whether every such coloring must contain at least one pair of equally colored neighbors. We prove that the answer is **no**: for every $n > 1$ there is a valid coloring in which *no* two neighbors share a color. Hence the proposed statement is **disproved**. The construction is a parity (checkerboard) coloring, and the underlying reason is that the grid graph of a face is bipartite, hence properly 2-colorable.

1 The problem, made precise

A physical $n \times n \times n$ twisty cube presents 6 faces to the eye. We model each face as an $n \times n$ array of *facelets* (the individual colored stickers), so the visible surface is a set of $6n^2$ facelets. A standard cube uses exactly six colors, and on the full surface each color occurs exactly n^2 times (one face's worth).

Definition 1 (Facelet positions)**.** Index the faces by $F \in \{1, \dots, 6\}$ and the cells of each face by (i, j) with $1 \le i, j \le n$. A *facelet* is a triple (F, i, j), and the set of all facelets is

$$V \;=\; \{\, (F, i, j) : 1 \le F \le 6,\ 1 \le i, j \le n \,\}, \qquad |V| = 6n^2.$$

Definition 2 (Within-face orthogonal adjacency)**.** Two distinct facelets (F, i, j) and (F', i', j') are *neighbors* iff they lie on the same face ($F = F'$) and are orthogonally adjacent in that face's grid, i.e.

$$|i - i'| + |j - j'| = 1.$$

This is exactly the "left/right/up/down" relation; diagonal cells and facelets meeting across a cube edge are *not* neighbors. Write $G = (V, E)$ for the resulting *adjacency graph*; it is the disjoint union of six copies of the $n \times n$ *grid graph* $P_n \,\square\, P_n$.[1]

Definition 3 (Valid cube coloring)**.** A *valid coloring* is a map $c : V \to \{1, \dots, 6\}$ assigning one of six colors to each facelet such that each color is used exactly n^2 times:

$$\left|\, c^{-1}(k) \,\right| = n^2 \qquad \text{for every color } k \in \{1, \dots, 6\}.$$

A pair of neighbors $(u, v) \in E$ is *monochromatic* if $c(u) = c(v)$.

[1] Here P_n is the path on n vertices and $\square$ denotes the Cartesian product of graphs; $P_n \,\square\, P_n$ is the standard $n \times n$ grid graph. See [3, 1].

With this language, the claim to be examined is:

> *(Claim.) For every $n > 1$ and every valid coloring of the $n \times n \times n$ cube, there exist at least two facelets of the same color that are neighbors.*

Equivalently: "every valid coloring has a monochromatic edge." We show this fails.

2 The key fact: a face is bipartite

Lemma 1 (Checkerboard parity)**.** *For every $n \geq 1$ the $n \times n$ grid graph is bipartite, with the two sides given by the checkerboard parity classes*

$$B = \{(i,j) : i+j \text{ even}\}, \qquad W = \{(i,j) : i+j \text{ odd}\}.$$

Consequently it is properly 2-colorable: the assignment $(i,j) \mapsto (i+j) \bmod 2$ gives no two orthogonally adjacent cells the same value.

Proof. Any edge of the grid joins (i,j) to (i',j') with $|i-i'|+|j-j'| = 1$, so exactly one coordinate changes by 1 and the parity of $i+j$ flips. Hence every edge joins B to W, so (B,W) is a bipartition and the parity map is a proper 2-coloring. (That grid graphs are bipartite is classical; see [1, 2].) □

Lemma 2 (Class sizes)**.** *In an $n \times n$ grid the two parity classes have sizes*

$$|B| = \left\lceil \tfrac{n^2}{2} \right\rceil, \qquad |W| = \left\lfloor \tfrac{n^2}{2} \right\rfloor, \qquad \text{so} \qquad |B| + |W| = n^2.$$

In particular, if n is even then $|B| = |W| = n^2/2$.

Proof. The cells (i,j) with $i+j$ even are counted directly: in each row the parities alternate, so a row contributes $\lceil n/2 \rceil$ or $\lfloor n/2 \rfloor$ cells to B depending on the row's starting parity, and summing over the n rows yields $\lceil n^2/2 \rceil$ in total; the complement W has the remaining $\lfloor n^2/2 \rfloor$. The two values sum to n^2 since $\lceil x \rceil + \lfloor x \rfloor = 2x$ for $x = n^2/2$ a half-integer or integer. □

3 Disproof of the Claim

Theorem 1 (Counterexample for all $n > 1$)**.** *For every integer $n > 1$ there exists a valid coloring $c : V \to \{1, \ldots, 6\}$ of the $n \times n \times n$ cube (six colors, each used exactly n^2 times) in which* no *pair of neighbors is monochromatic. Therefore the Claim is false.*

Proof. Partition the six faces into three pairs,

$$\{F_1, F_2\}, \quad \{F_3, F_4\}, \quad \{F_5, F_6\},$$

and assign to the three pairs the three color-pairs

$$\{1,2\}, \qquad \{3,4\}, \qquad \{5,6\},$$

respectively. It suffices to describe the coloring on one pair of faces with a given color pair $\{a,b\}$; the other two pairs are identical up to renaming colors.

Let B and W be the parity classes of Lemma 1. On the first face of the pair, color

$$c(i,j) = \begin{cases} a, & (i,j) \in B, \\ b, & (i,j) \in W, \end{cases}$$

and on the second face of the pair, use the *opposite* assignment

$$c(i,j) = \begin{cases} b, & (i,j) \in B, \\ a, & (i,j) \in W. \end{cases}$$

No monochromatic neighbors. Every edge of G lies inside a single face. On either face the coloring is a function of the parity $(i+j) \bmod 2$ alone, and by Lemma 1 adjacent cells have opposite parity, hence opposite colors. So no edge is monochromatic. (This already holds on each face in isolation, so it holds on the whole cube, whose adjacency graph is the disjoint union of the six faces.)

Each color is used exactly n^2 times. Consider color a in the pair $\{a, b\}$. On the first face it occupies B, i.e. $\lceil n^2/2 \rceil$ cells; on the second face it occupies W, i.e. $\lfloor n^2/2 \rfloor$ cells. By Lemma 2,

$$\#\{a\} = \left\lceil \tfrac{n^2}{2} \right\rceil + \left\lfloor \tfrac{n^2}{2} \right\rfloor = n^2.$$

By symmetry color b also appears $\lfloor n^2/2 \rfloor + \lceil n^2/2 \rceil = n^2$ times. The same holds for colors $3, 4$ and $5, 6$ on their respective pairs. Thus all six colors occur exactly n^2 times, so c is a valid coloring.

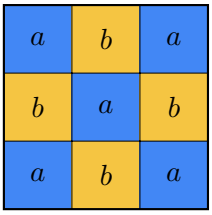
We have exhibited a valid coloring with zero monochromatic neighbors, for every $n > 1$ (indeed for every $n \geq 1$). This contradicts the Claim, which asserts at least one monochromatic neighbor must always exist. Hence the Claim is disproved. □

Remark 1 (The "swap" is what balances the inventory)**.** A single global 2-coloring (one checkerboard, two colors) would also have no monochromatic edge, but it would use two colors $3n^2$ times each and the other four not at all, violating the sticker inventory. Splitting into three color-pairs and swapping $a \leftrightarrow b$ between the two faces of a pair is precisely what restores the exact count n^2 per color while preserving the proper-coloring property. For even n the swap is unnecessary for *counting* (each class already has $n^2/2$ cells), but it does no harm; for odd n it is essential, since one class is larger than the other.

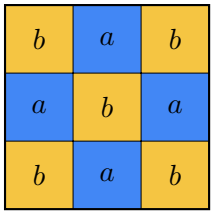
Remark 2 (It is not a near miss: zero adjacencies)**.** The construction achieves not merely "few" but *zero* monochromatic neighbors. Equivalently, the proper chromatic number of each face is 2 (Lemma 1), so two colors per face already suffice to avoid all same-color adjacencies, and we have six colors to spend. The constraint "exactly n^2 of each color" is the only thing that makes the problem nontrivial, and the pairing argument dispatches it.

4 Worked example: the ordinary $n = 3$ cube

For $n = 3$ each face is a 3×3 grid with $n^2 = 9$ cells. The parity classes have sizes $|B| = \lceil 9/2 \rceil = 5$ and $|W| = \lfloor 9/2 \rfloor = 4$. Take the color pair $\{a, b\}$ on two opposite faces:



Face 1



Face 2

Face 1 uses a five times and b four times; Face 2 uses b five times and a four times. Across the pair, a appears $5+4=9$ times and b appears $4+5=9$ times — exactly n^2 each. On each face the two colors alternate like a checkerboard, so no two orthogonally adjacent cells match. Repeating with color pairs $\{3,4\}$ and $\{5,6\}$ on the remaining two face-pairs completes a valid $3\times3\times3$ coloring with no monochromatic neighbor whatsoever.

5 Discussion and scope

What was and was not assumed. We treated the cube as a pure stickering problem: six colors, each used exactly n^2 times, placed on the six faces in *any* arrangement, with adjacency taken only *within* a face. Under these (faithful) rules the Claim is false. Two different readings would change the problem:

1. *Across-edge adjacency.* If facelets meeting across a shared cube edge also counted as neighbors, the adjacency graph would gain edges linking the six faces, and one would need to check that a global 2-coloring still exists; the counting argument would require more care. (This was explicitly excluded here.)

2. *Only legal scrambles.* If one restricts to states reachable from the solved cube by legal moves (fixed centers for odd n, permutation and orientation parity constraints, etc.), the set of admissible colorings shrinks dramatically and the question becomes a genuinely different — and much harder — combinatorial problem on the cube group, not resolved by the elementary argument above.

For the "any valid stickering" reading requested here, Theorem 1 settles the matter completely.

Summary. The statement "an $n\times n\times n$ cube ($n>1$) always has two same-colored orthogonal neighbors on some face" is **false**. Because each face is a bipartite grid graph, a checkerboard coloring avoids all same-color adjacencies, and pairing the six faces into three two-color checkerboards (swapping the two colors between paired faces) keeps every color's count at exactly n^2. This yields, for every $n>1$, a valid coloring with zero monochromatic neighbors.

References

[1] R. Diestel, *Graph Theory*, 5th ed., Graduate Texts in Mathematics 173, Springer, 2017. (Bipartite graphs and 2-colorability: a graph is bipartite iff it has no odd cycle; grid graphs are bipartite.)

[2] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001. (Proper colorings, chromatic number, $\chi(G)=2$ for bipartite G.)

[3] F. Harary, *Graph Theory*, Addison-Wesley, 1969. (Cartesian product of graphs and the grid graph $P_m \,\square\, P_n$.)